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Sound-based navigation system for visually impaired individuals

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ABSTRACT

Millions of people worldwide have impaired vision for various reasons, such as cataracts, glaucoma, AMD, diabetic retinopathy, trauma, infections, genetic disorders, and amblyopia. Visual impairment ranges from normal vision to complete blindness, with varying severities that affect daily life. This study presents a sound-based navigation system to improve indoor mobility for visually impaired individuals. The system combines voice recognition, LED-guided floor paths, and real-time haptic feedback to help users follow predefined routes. Users issue voice commands to activate illuminated floor strips, which a robot cart follows using light-dependent resistor (LDR) sensors. An electromechanical compass provides accurate orientation, while a vibrating handle delivers haptic alerts for real-time directional adjustments. Bluetooth Low Energy (BLE) determines the user's location and destination to guide them along the correct route. Experimental results show high accuracy: 91.04 % voice recognition, 4 % navigation error due to sensor noise, and 3 % speed regulation error from electrical fluctuations. The proposed solution is cost-effective and user-friendly, promoting greater independence and safety for visually impaired individuals.

1. Introduction

Recent estimates indicate that about 43.3 million people worldwide are blind, and another 295 million experience moderate to severe vision impairment (Vision Loss Expert Group of the Global Burden of Disease Study, 2024). These numbers highlight visual impairment as a significant global public health concern. Blindness arises from many causes. Cataracts, involving clouding of the eye's lens, are often age-related. Glaucoma includes a group of conditions that damage the optic nerve, usually due to high eye pressure. Age-related Macular Degeneration (AMD) causes loss of central vision and accounts for 6–9 % of global blindness cases (Fleckenstein et al., 2021). Diabetic Retinopathy damages the retina's blood vessels due to diabetes (Wang & Lo, 2018). Physical trauma can also lead to blindness (Stathopoulos et al., 2018). Infections such as trachoma and ocular herpes may cause severe vision loss (Alemayehu & Cherinet, 2019). Genetic disorders like retinitis pigmentosa result in gradual vision decline (Cross et al., 2022). Amblyopia, or "lazy eye," occurs when one eye fails to develop proper vision (Kolhe & Diaghvane, 2021). Visual impairment is classified by severity (Bourne et al., 2017): Normal Vision: Clear vision with no impairment; Mild Visual Impairment: Minor vision loss with little effect on daily life; Moderate Visual Impairment: Visual acuity below 20/70 but better than 20/200; Severe Visual Impairment: Visual acuity below 20/200 but better than 20/400; Profound Visual Impairment: Visual acuity below 20/400 with some light perception; Total Blindness: No light perception at all. Various technologies assist visually impaired individuals. Screen Readers convert on-screen text to speech for digital access (Kuber et al., 2012). Braille Displays translate text into Braille for tactile reading (Rantala et al., 2009). Smart Glasses use cameras and AI to describe surroundings through audio (Mukhiddinov & Cho, 2021). GPS and Navigation Apps provide spoken directions for safe travel (Tyagi et al., 2021). Tactile Maps and Graphics offer raised diagrams for spatial understanding (Mukhiddinov & Kim, 2021). Voice-activated assistants like Amazon Alexa and Google Assistant perform tasks via voice commands (Almurayziq et al., 2022). Mobile Apps support object identification, navigation, and reading assistance (Christy & Pillai,

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2021). Advanced techniques now aid both indoor and outdoor navigation. One promising area is AI-powered obstacle detection and avoidance, using deep learning and LiDAR sensors for real-time perception and path planning (Krupáš et al., 2024). These systems detect and classify obstacles, allowing dynamic route adjustments in complex environments. Wearable vision aids, such as smart glasses and sensory belts, integrate cameras, microcontrollers, and auditory or haptic feedback to interpret the surroundings. They rely on computer vision to recognize objects and motion, translating this data into vibrations or synthesized speech (Tokmurziyev et al., 2024). Computer vision-based navigation systems use object detection and simultaneous localization and mapping (SLAM) to create indoor maps and track user positions (Singh et al., 2022). These systems employ monocular or stereo vision to provide directional cues based on visual features. Recent Edge AI developments have enabled low-power wearable devices to process navigation tasks locally, reducing reliance on cloud infrastructure. This improves response time, privacy, and reliability (Özkan and Şahin). AI-powered mobile applications enhance geospatial awareness through voice assistants that provide auditory directions, location queries, and natural language interaction (Nama, 2023). Finally, multimodal sensor fusion has emerged as a powerful approach to localization. It combines data from GPS, Bluetooth Low Energy (BLE), ultrasonic sensors, and computer vision to improve accuracy indoors, where GPS signals are weak (Fischer et al., 2022). These hybrid systems enhance precision and reliability by compensating for individual sensor limitations.

2. Materials and methods

2.1. The method

This study presents an intelligent audio-navigation system designed to assist visually impaired individuals in navigating indoor environments. The system features a robot car that autonomously follows an LED strip lighting system installed on the floor (Fig. 1). Users control navigation via voice commands (e.g., saying "Room 2″), which activate the corresponding light track via a relay array.

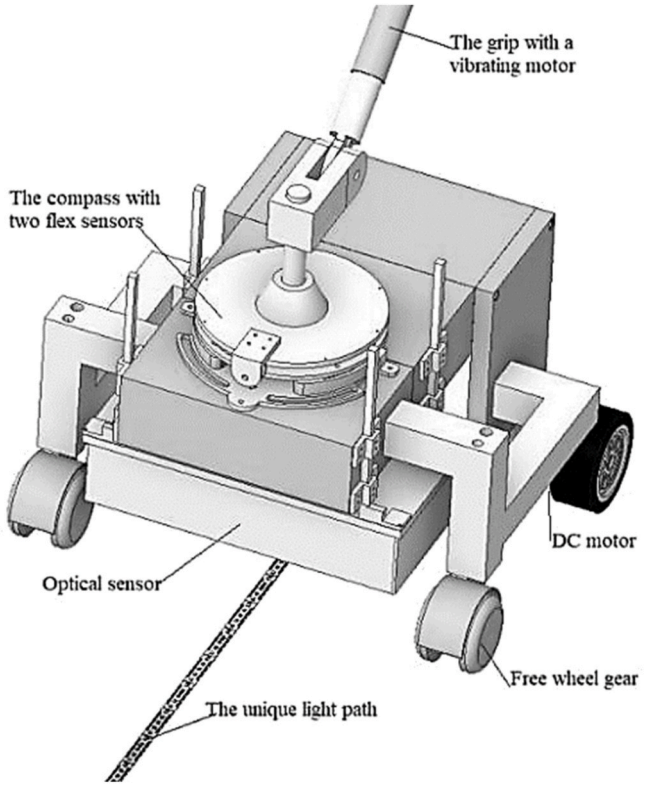


Fig. 1. 3D model of the Proposed Robot Car for Impaired Individuals Indoor Navigation.

2.1.1. The components and functionality

1. Light Sensor Navigation System: The robot car has a light sensor circuit that detects the LED strip intensity. A microcontroller processes this data and adjusts the motor driver to follow the illuminated path.
2. Electro-Mechanical Compass: This device aligns the user and vehicle to the light path and controls the car's speed. The compass detects deviations between the user's and the vehicle's position.
3. Haptic Feedback via Vibrating Handle: The user holds a rigid handle with three vibrating motors (near the thumb, index, and little finger) that provide tactile alerts for directional corrections (forward, right, left). Vibration intensity reflects misalignment.
4. Bluetooth Low Energy (BLE) for Positioning: BLE technology determines the user's indoor location and destination. The system supports five predefined destinations (Room 1-Room 5).

2.2. Working principle

This system provides intuitive, real-time indoor guidance for visually impaired individuals, as illustrated in the following steps (Fig. 2):

1. Voice Command Input (Destination):

The user issues a voice command through a Bluetooth headset. The microcontroller recognizes the command and identifies the destination by matching a predefined keyword.

2. Activation of the Unique Light Path:

Using voice command and Bluetooth, the system determines the user's current position and destination (hot keyword). When identified, it activates the specific floor light path via relay matrices. Only the particular light path leading to the selected destination will be activated.

3. Robot Car Navigation System:

Once the unique light path is activated, the optical sensor detects the light's alignment (left, center, right). The microcontroller adjusts the wheel driver to ensure the robot car accurately follows the illuminated path.

4. Mechanical Compass and User Alignment Monitoring System:

The robot car features a mechanical compass with two flex sensors embedded in its body. On a straight path, both sensors maintain low resistance. If the car turns and the user misaligns, resistance increases on one side, indicating a deviation (left or right). The microcontroller processes this, adjusts the car's speed, and alerts the user.

5. Vibration Motors (User Feedback):

The user holds a handle attached to the robot car with three vibrating motors for feedback (thumb, middle finger, and pinky). The corresponding motor vibrates based on the car's movement direction:

- Forward: Middle finger vibration
- Right turn: Thumb vibration
- Left turn: Pinky vibration

If misalignment occurs, the system slows or stops until the user repositions to match the car's direction.

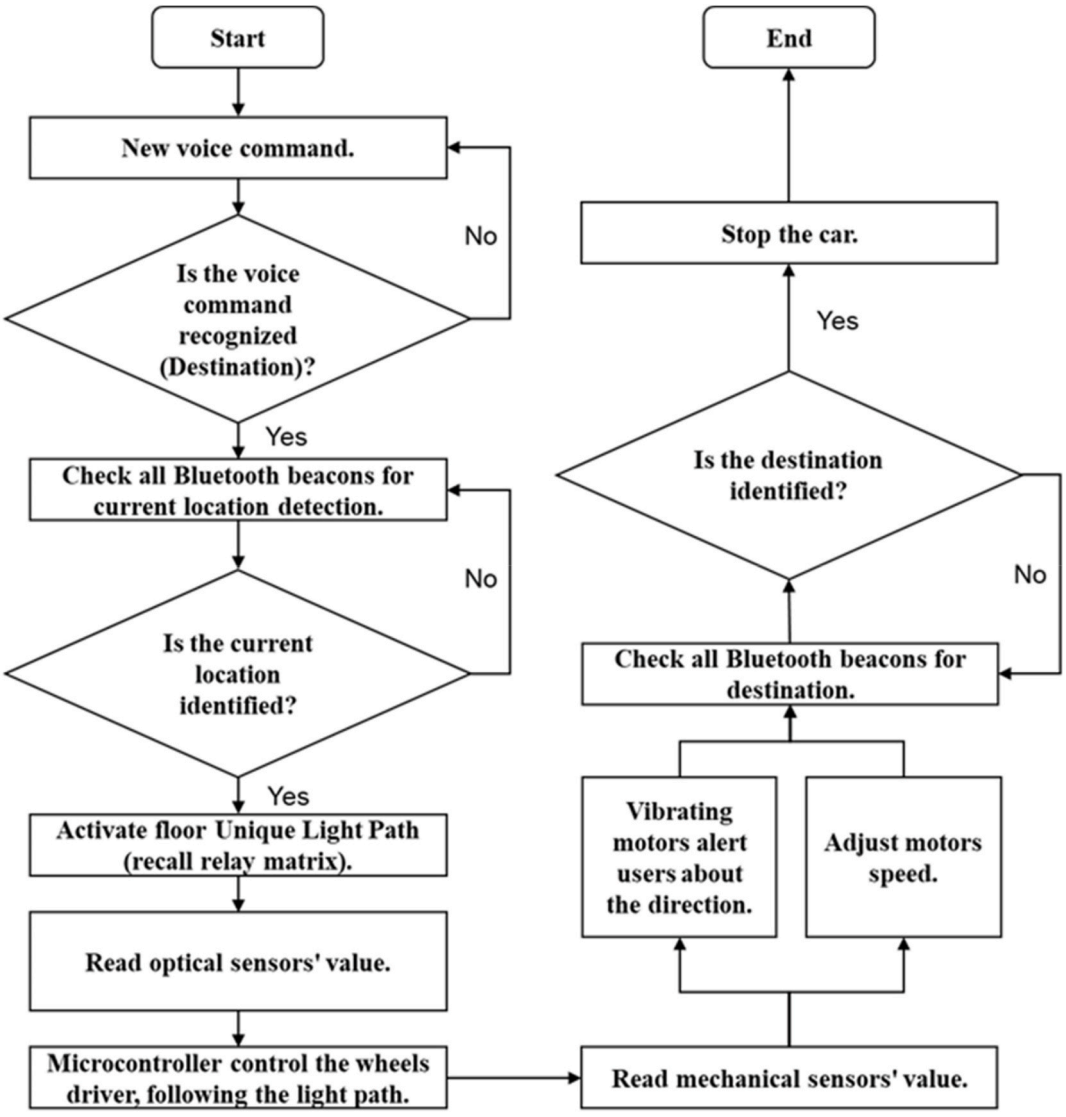


Fig. 2. The block diagram for the Intelligent Audio-Navigation System for Visually Impaired Individuals. It illustrates the system's parameters, including voice input, Bluetooth communication, LED light path, optical sensors, mechanical compass, motor driver, and destination detection.

6. Final Destination Recognition:

Bluetooth Low Energy (BLE) technology identifies both the start and the destination. Once the robot car reaches the designated location, it stops and waits for a new voice command to begin the next navigation cycle.

2.3. System architecture

2.3.1. The communications

Effective communication between the user and the system is essential for successful indoor navigation. Since the user is visually impaired, audio communication is the most suitable method. The study developed a simple communication scenario for the user and vehicle in an indoor environment (Fig. 3). The user issues destination commands via a Bluetooth microphone, and a neural network (Snowboy) processes the voice inputs to recognize specific keywords (Bakouri et al., 2023). The system is trained to identify five Arabic "hot keywords," each corresponding to a destination (Rooms 1–5). When a voice command is detected, the microprocessor activates the required relays to control the LED floor light path, guiding the vehicle along the illuminated route. Using a convolutional neural network, the system recognizes user commands based on extracted audio features, referred to as "hot keywords" (e.g., "Room 1″). The key features of this approach include the use of Mel-Frequency Cepstral Coefficients (MFCCs) to extract feature maps from audio files (Fig. 4), providing an adequate representation of speech signals. On the other hand, a Finite Impulse Response (FIR) filter enhances signal processing. The Snowboy algorithm then learns and recognizes patterns in these extracted features.

2.3.2. The user's location and destination

This study builds on previous work using the nearest neighbor recognition technique with Bluetooth Low Energy (BLE) (Alfahad et al., 2025) to locate visually impaired users indoors, achieving an average error of 0.5–1 m. This method accurately determines the user's current location and the room they are in. When the user reaches the destination, the system issues an alert and resets the navigation cycle (Fig. 5). The internal map is based on a pre-programmed relay matrix (Table 1), which activates specific relays embedded in the floor. A Python script determines which relays to activate for each route. The monitoring system employs a Raspberry Pi as the master unit and BLE nodes as slave units to collect data and calculate positioning from signal strength within a 10 × 10-m indoor space. The BLE nodes periodically broadcast signals before entering sleep mode. A Received Signal Strength Indicator

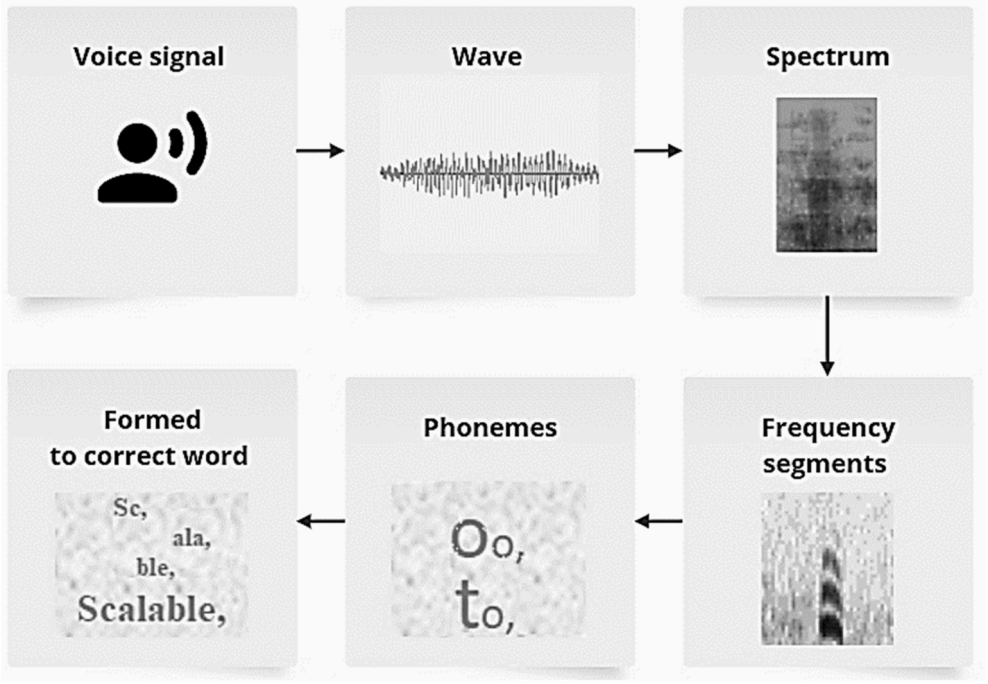


Fig. 3. The flowchart shows the main processing steps for voice recognition.

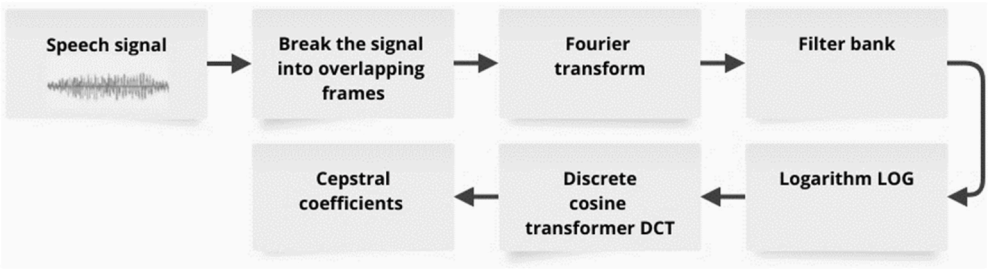


Fig. 4. Structure of the MFCC model.

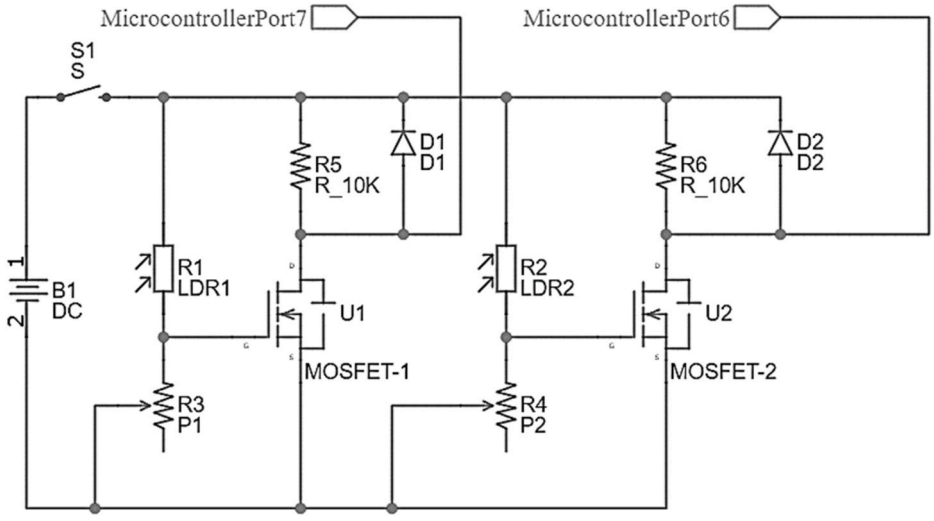


Fig. 5. Complete circuit diagram of the light sensor module used to trigger the microcontroller and activate the robot car's light-following functionality.

(RSSI) and an auto-adjusting algorithm improve localization accuracy. When a smart device enters the broadcast area, it detects the access point with the strongest signal. Anchor-based algorithms use RSSI to estimate distances efficiently. The proposed scaling algorithm is hardware-efficient, with low computational complexity, and maintains reliable performance with few nodes, achieving an average positioning

Table 1
Directional map for five trips.

Current location	Destination	Relays Pins								
		1	2	3	4	5	6	7	8	9
Room-1 to	Room-2	✓	x	✓	✓	x	x	x	x	x
Room-1 to	Room-3	✓	✓	x	x	x	x	x	x	x
Room-1 to	Room-4	✓	x	✓	x	✓	✓	x	x	x
Room-1 to	Room 5	✓	x	✓	x	✓	x	✓	x	x
Room-1 to	Room-6	✓	x	✓	x	✓	x	✓	x	x
Room-2 to	Room-3	x	x	✓	✓	x	x	x	x	x
Room-2 to	Room-4	x	x	x	✓	✓	✓	x	x	x
Room-2 to	Room 5	x	x	x	✓	✓	x	✓	✓	x
Room-2 to	Room-6	x	x	x	✓	✓	x	✓	x	✓
Room-3 to	Room-4	x	✓	✓	x	✓	✓	x	x	x
Room-3 to	Room 5	x	✓	✓	x	✓	x	✓	✓	x
Room-3 to	Room-6	x	✓	✓	x	✓	x	✓	x	✓
Room-4 to	Room 5	x	x	x	x	x	✓	✓	✓	x
Room 4 to	Room-6	x	x	x	x	x	✓	✓	x	✓
Room 5 to	Room-6	x	x	x	x	x	✓	x	✓	✓

error of 0.5–1 m. The Raspberry Pi and Bluetooth peripherals (HC-05 modules) communicate within a 10 × 10-m range via the Serial Port Protocol (SPP). Multiple slave units connect simultaneously, and the microprocessor identifies the strongest signal to determine the user's location. The system stores five room identifiers, each linked to a unique MAC address. RSSI values estimate distance by measuring radio signal strength, expressed in decibels (dBm), where zero dBm represents the maximum signal and increasingly negative values indicate weaker signal strength. The command structure retrieves RSSI values while discarding connections with RSSI values above −20 dBm to reduce inaccuracies.

2.3.3. Activation of unique light path (relays matrix)

The system's operating principle is simple yet efficient: the light path is activated by two primary inputs. First, the user's current position is determined using Bluetooth detection within the designated area. Second, the destination is identified via a user-issued voice command. Only after receiving both inputs does the system reference the relay matrix, as shown in Table 1. Once this data is processed, the corresponding light paths are triggered to guide the robot car toward the selected destination. Each route may consist of multiple light strips and directional turns, depending on the indoor layout and the required endpoint. The relay matrix ensures that only the necessary relays are activated, minimizing energy use and improving navigation accuracy.

2.3.4. The light sensors

A light sensor circuit is designed as a voltage divider, using a Light Dependent Resistor (LDR) and a potentiometer to adjust the output voltage (V sub out) based on light intensity (Fig. 5). The circuit operates according to Ohm's law, applying the voltage divider formula to regulate a MOSFET switch. This configuration sends light intensity data to the microcontroller and controls the robot's DC motors. The circuit includes several essential components: Light Dependent Resistor (LDRs) that sense light levels 10 *omega resistors* r adjusting sensitivity, N-channel MOSFETs serving as electronic switches, and DC motors that drive the robot based on light direction (Table 2). The operational process is straightforward. Each photoresistor forms a voltage divider with a potentiometer. When light strikes one side, the corresponding LDR's resistance decreases, increasing the voltage Vsubout. his voltage is then applied to the MOSFET's gate terminal. If the output voltage from the bridge circuit surpasses the MOSFET's threshold voltage, the MOSFET activates, allowing current to pass to the relevant microcontroller input. When both sensors receive equal light intensity, both motors are energized, causing the robot to move forward in a straight line.

Table 2
Microcontroller logical map for robot car movement.

Light Detected at	Relay Pins Activated				Wheel Movement		Robot Movement Direction
	1	2	3	4	Right	Left	
Left sensor	✓	x	x	x	✓	x	To the left
Right sensor	x	x	✓	x	x	✓	To the right
Both Sensors	✓	x	✓	x	✓	✓	Forward
No Light Detected	x	x	x	x	x	x	Stop

Fig. 6 shows how the light sensor controls the output voltage. As light levels increase, the LDR's resistance decreases, raising the MOSFET's gate voltage. Once this voltage (V_GS) exceeds the threshold (about 2.5 V), the MOSFET turns on and allows current to flow to the microcontroller pins. The connected motor then runs based on the control logic shown in Table 2. At low light levels, the LDR has high resistance, so the gate voltage stays low and the MOSFET stays off. This creates a light-controlled switch that activates the motors only when there is sufficient light, keeping the tracking system accurate and efficient, as shown in Fig. 6.

2.3.5. Robot car movements

The robot car's motion is managed by a microcontroller that processes light-sensor data and commands a driver motor, translating input measurements into appropriate propulsion actions. The platform features four wheels, with the rear wheels (left and right) powered by 12-V, 50-W N20 DC gear motors. They are controlled by a quad-relay array operated with a negative trigger signal. The front wheels are non-powered to permit passive steering and free movement. The proposed method directs the robot to follow a light path activated by the microcontroller. Since the light path, the ground, and the sensors are positioned perpendicular to each other and very close, shadows are cast at the measurement site. This effectively blocks ambient background light, enhancing measurement accuracy. Regarding the robot's electrical circuit, the front wheels are fully mobile, while the rear wheels are powered by a 12V DC power source, with 80 % motor efficiency and 3 Nm of torque at full load. Movement is controlled based on light intensity detected by two optical sensors, as shown in Table 2. The relay control logic operates as follows:

- when the left sensor reports greater light intensity than the right, pin 1 of the quad relay is activated;
- when the right sensor dominates, pin three is activated;
- and when both sides read equal light, pins 1 and 3 are activated simultaneously.

Through experimental evaluations, a minimum light intensity threshold of 1.5 lux was established. During navigation, the Bluetooth network distributed throughout the indoor environment identifies the user's current location. Thus, the starting point is automatically detected while the user provides the destination via voice command. The Bluetooth system continuously monitors navigation accuracy. When the user reaches the destination, the system detects the corresponding Bluetooth signal and confirms the trip's successful completion.

2.3.6. The user and robot alignment

In this work, the visually impaired user follows a robot cart using a guiding stick, analogous to traditional guide-dog assistance. The cart advances while the stick is pulled behind it, enabling the user to follow the cart's direction. Vibration motors embedded in the stick's handle notify the user of directional changes. However, when lateral deviations of the cart can leave the user uncertain of its position, an alert and alignment system is essential. To address this, an electromechanical compass (Fig. 7) provides continuous, real-time alignment between the user's motion and the robot's trajectory. In this way indoor navigation accuracy improves by continuously correcting the user's path.

The compass comprises three principal components:

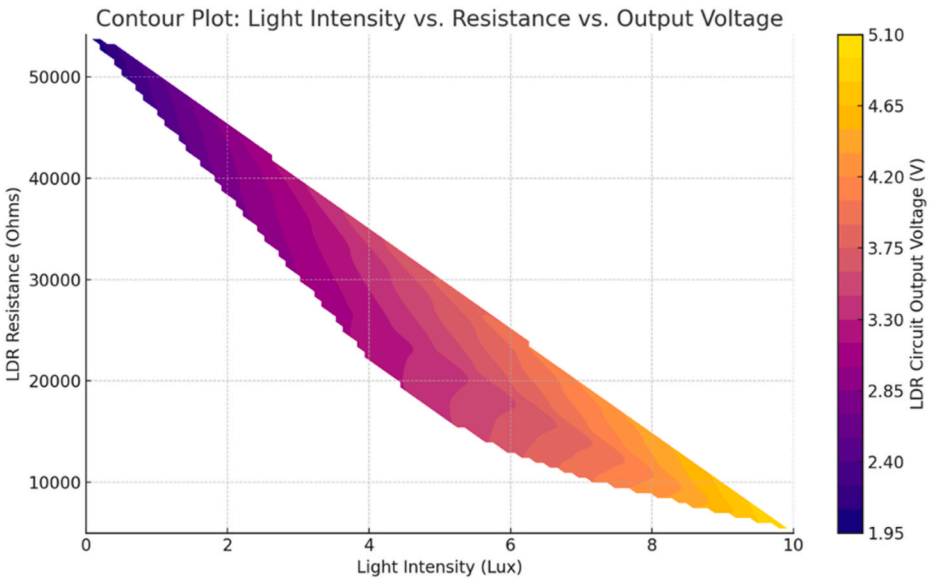


Fig. 6. Contour plot illustrating the relationship between light intensity and LDR resistance, along with the corresponding MOSFET switching status. The color gradient transitions from purple (MOSFET OFF) to yellow (MOSFET ON), activation threshold (color value 2.8).

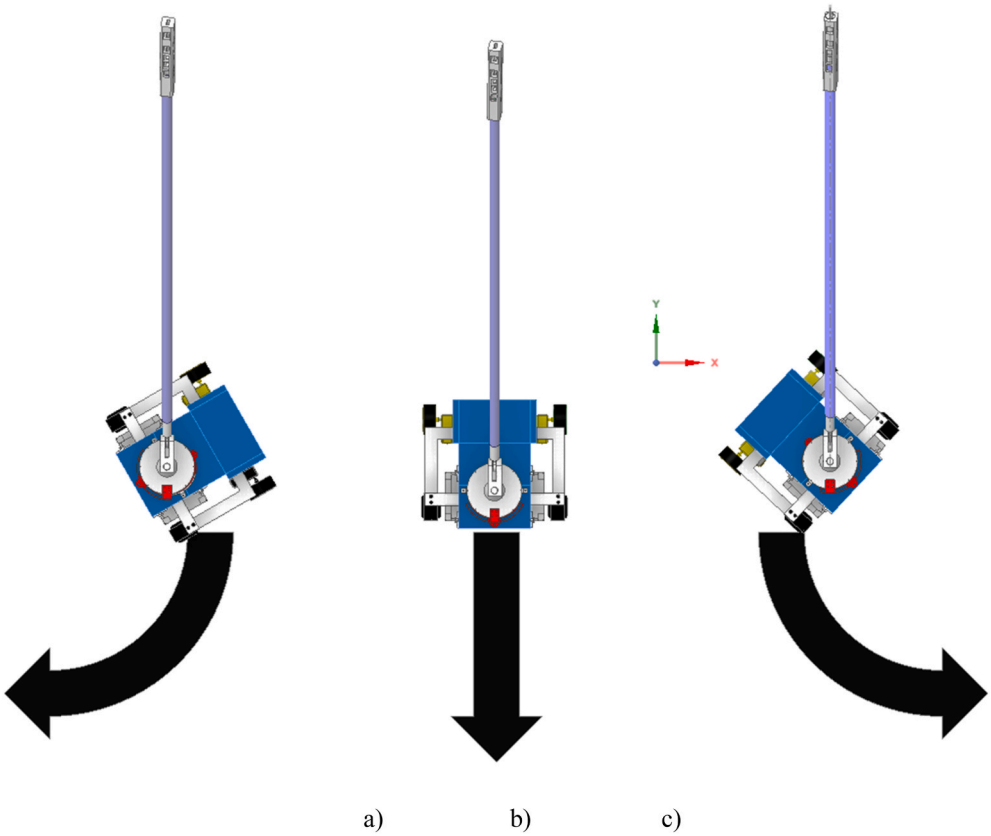


Fig. 7. A diagram illustrating how the robot responds to the user's input. The figure shows three different positions of alignment between the upper and lower parts of the compass caused by the car drifting and user alignments: a) The car to the right, user forward; b) The car forward, user forward; c) The car to the left, user forward.

- a fixed lower part rigidly mounted on the robot to ensure it follows the robot's heading;
- a hinged, movable upper part that rotates with minimal friction in response to torque applied through the user's stick and returns to neutral via a torsion spring when no force is present, thus realigning the user with the robot's direction;
- and the user stick itself, a hollow aluminum tube affixed to the upper part that transmits electrical signals between the handle and the microcontroller to enable real-time feedback and control.

Two flex sensors housed within the compass body—one head fixed to the upper part and the back anchored to the lower part—detect

misalignment. These sensors detect deviations in user movement relative to the robot's direction. When the user moves in perfect alignment with the cart, sensor readings remain balanced. If the user veers left or right, the applied torque causes the upper compass layer to rotate, changing the flex sensor resistance. While torque-induced rotation of the upper layer alters sensor resistance, the microcontroller interprets the resulting resistance change to quantify the deviation and trigger corrective measures.

To convey corrective instructions, three vibrating motors are embedded in the handle.

- o Thumb vibration → signals the user for a proper correction.
- o Middle finger vibration → signals the user for a forward alignment.
- o Pinky finger vibration → signals the user for a left correction.

If the user deviates from the robot's path, the system will:

- o Alert the user through directional vibrations to guide them on the correct path.
- o Adjust the robot's speed—slowing down or stopping—if misalignment continues.
- o Resume movement once alignment is restored, guiding the user to the target destination.

Navigation continues until the robot detects the Bluetooth signal corresponding to the designated location. At that point, it stops and waits for the following voice command.

In cases of extreme deviation (e.g., a 90° offset), the system notably slows and ultimately stops the vehicle, alerts the user to realign, and resumes only once proper alignment is achieved (Fig. 8).

2.3.7. Car robot speed calculation

A simple circuit was built to track changes in resistance from the flex sensors and control the robot car's speed (Fig. 10a). The servo motor angle changes according to the flex sensor readings. The servo is mechanically connected to a 200Ω potentiometer, which is wired in series with the power supply feeding the robot's motors. When the servo is at 0°, it allows maximum current to the motors; at 180°, the current drops to zero, stopping the motors. The circuit incorporates the following key components:

- o The mechanical compass contains two flex sensors with a resistance range of 30–136 $k\Omega$, capable of detecting bending angles from 0 to 180° As the user and car align (Fig. 9a).
- o The microcontroller translates the flex-sensor readings into servo rotation angles within the 0° and 180° Ange.
- o The servo motors are mechanically coupled to a zero-200Ω potentiometer that adjusts the car motor current from 0 to 2500 mA.

Minor bending of the flex sensor produces only a negligible effect on the robot's speed (Fig. 9b), whereas the maximum resistance corresponds to the slowest movement. Fig. 9a presents the prototype model with the groove bend sensor, and Fig. 10a depicts the electronic circuit that converts flex-sensor resistance values into servo-motor rotations. Fig. 9b shows the relationship between increasing flex-sensor resistance and bending angle. In contrast, Fig. 10b demonstrates how the servo angle adjusts accordingly to map flex-sensor resistance to car-speed control.

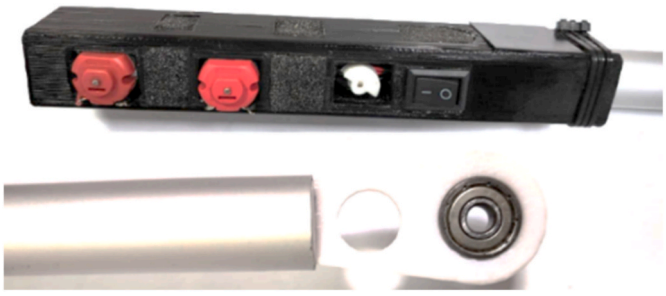


Fig. 8. Fig. 8. The user stick is equipped with three vibration feedback motors for directing the user. The design is inspired by the vibration feedback mechanism commonly used in mobile devices.

3. Results

3.1. Evaluation of voice recognition algorithm

Voice communication enables visually impaired individuals to operate devices and receive messages from intelligent systems. It also allows them to interact with their environment and society, supporting independent management of their personal lives. The user interacts with the robotic car through a simple voice scenario:

- Successive orders: Each action follows the completion of the previous one, and the next command is executed only after the current action is finished.

Five trained Arabic voice commands were employed to represent five destinations, and the system incorporated a biometric feature. The evaluation involved 35 commands from trained users and 32 from untrained users (Table 3). Voice recognition accuracy was assessed by the robot's ability to respond correctly to commands, and a confusion matrix (Fig. 11) compared correctly and incorrectly classified voices against the actual targets for both groups (35 commands for trained users and 32 for untrained users). The experimental data yielded measures of True Positives (TP), False Positives (FP), True Negatives (TN), and False Negatives (FN), as presented in Table 3.

The resulting performance metrics were: Accuracy 91.04 %, Precision 97.14 %, Recall (Sensitivity) 87.18 %, and F1 Score 91.89 %.

3.2. Verification of the Cart's navigation

The experiment assesses the robot cart's ability to follow a predefined light path, including forward movement and right- and left-hand turns, over 3 min. The setup features a robotic cart equipped with LDR-based light sensors that navigate a zig-zag light path laid on the floor, while a data logging system records sensor readings and movement patterns. The cart traverses the path by detecting variations in light intensity, with deviations and actual path data analyzed post hoc to determine trajectory accuracy. The LDR detects changes in light intensity and converts them into an electrical signal, which is processed by a voltage divider composed of a fixed 50 $K\Omega$. The resulting voltage governs whether the N-channel MOSFET switches on or off, enabling the microcontroller to control wheel actuation. Specifically, when light intensity exceeds 1.5 lux, the LDR resistance drops to 10 $k\Omega$, raising the MOSFET gate voltage above 2.5 V turning the device on to activate the wheels; conversely, when light intensity falls below 1 lux, the LDR resistance rises to 50 $k\Omega$, Lowering the MOSFET gate voltage and keeping the wheels off (Fig. 12). Results show that the robot closely tracks the light path, with minor deviations primarily due to background noise that affects sensor readings. The observed path adherence yields a 4 % error rate, attributed to small voltage fluctuations from electrical noise and sensor variability (Equation (1)). The percentage error between the observed and expected values was computed, and a slight discrepancy was noted between the expected and actual Gate-Source voltage for the transistor. The following calculation was used:

$$Percentage\ error = \frac{|V_{observed} - V_{true}|}{V_{true}} \times 100\% \quad (1)$$

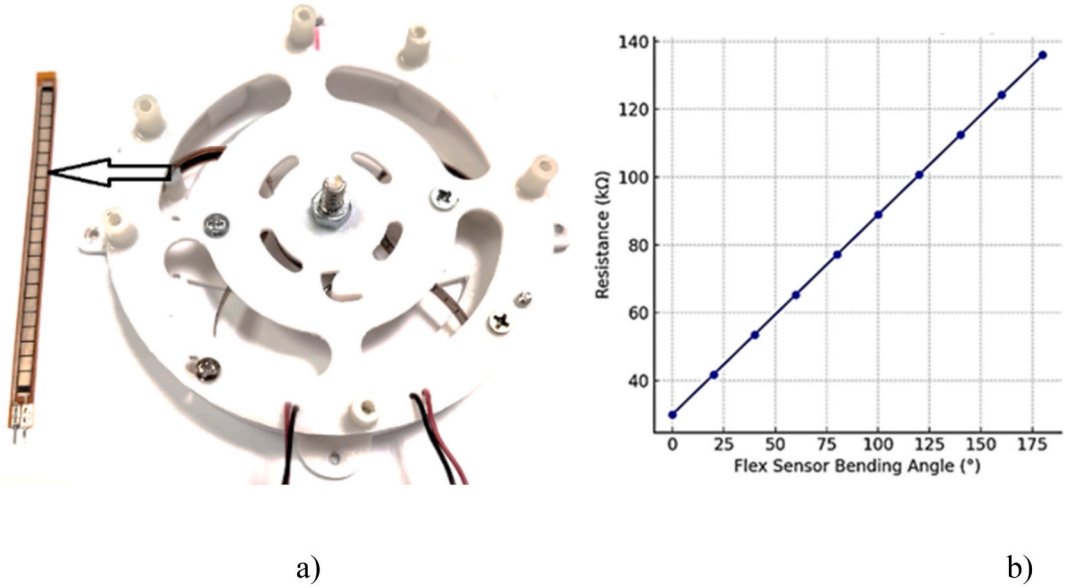


Fig. 9. Mechanical Compass Integrated with Flex Sensors for Real-Time Feedback
(a) Prototype Model: The mechanical compass features a movable upper plate and a fixed lower base, incorporating two flex sensors to monitor angular bending. These sensors provide real-time resistance data to the microcontroller for processing and analysis.
(b) Resistance vs. Bending Angle Graph: This graph illustrates the sensor response, where resistance increases from 30 kΩ to 136 kΩ as the bending angle ranges from 0° to 180°.

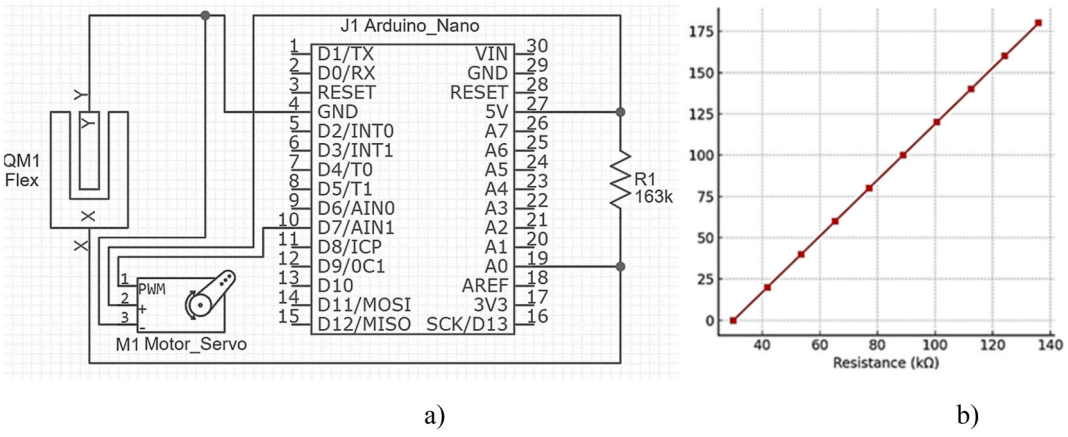


Fig. 10. Shows a microcontroller circuit converting flex sensor resistance into servo motor rotation that adjusts the potentiometer of the motor's current. Increased flex resistance reduces speed, ensuring real-time alignment and safe navigation between the user and robot:
(a) The microcontroller electronic circuit converts flex resistance into servo motor rotation, adjusts the motor current, and controls vehicle speed.
(b) Displays graphs showing the ideal relationship between flex resistance (30–136 kΩ) and servo angle (0°–180°).

Table 3
The Voice recognition experimental matrix.

Target value	Predicted Negative	Predicted Positive
Actual Negative	27 (TN)	1 (FP)
Actual Positive	5 (FN)	34 (TP)

$$Percentage\ error = \frac{|144.932163 - 139.3578509|}{139.3578509} \times 100\% \quad (2)$$

$$Percentage\ error = 3.99\%.$$

3.3. Robot car speed evaluation

The experiment assesses the user/robot alignment system by analyzing the readings from the mechanical compass and the microcontroller's regulation of the robot car's speed, including the activation of vibrating motors for user guidance. The system comprises flex sensors, servo motors, potentiometers, and a power-regulating circuit. The servo angle determines the car's speed through the appropriate motor current: at 0°, maximum current and speed are allowed, while at 180°, the motor is halted. The experiment monitors how flex-sensor readings drive the servo adjustment, which in turn adjusts the potentiometer and power supply to control the robot's movement. In the experimental graph (Fig. 13), the blue line represents speed variations as a function of flex-sensor input, illustrating the relationship between the bend angle and car speed. As the bend angle increases, the resistance rises, reducing the power delivered to the motor and decreasing the speed. The system achieves a maximum speed of 0.5 m/s at 0° bend and gradually slows to a stop as the flex angle approaches 190°.

A 3 % error in the primary power-line current (mA) is attributed to electrical fluctuations in the power supply and mechanical

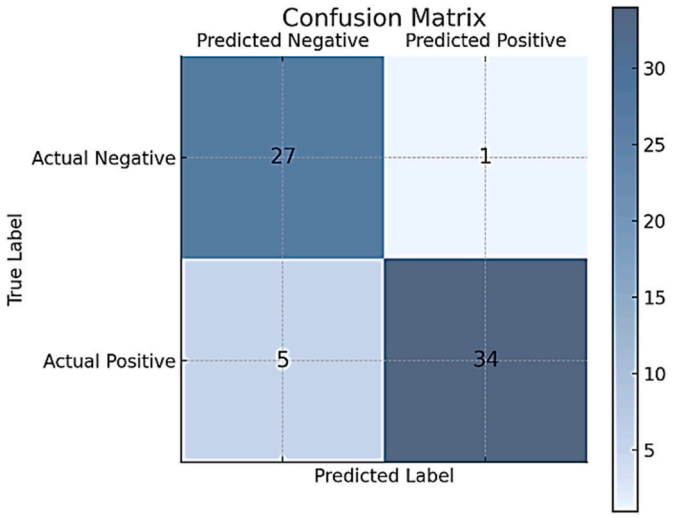


Fig. 11. The confusion matrix visualization for evaluating voice recognition accuracy.

inconsistencies in the servo motor that affect resistance feedback, with the percentage error between observed and expected motor-supply values computed (Equation (4)). A slight discrepancy was evident between the expected and actual motor-supply currents, as reflected in the accompanying calculation.

$$Percentage\ error = \frac{|I_{observed} - I_{true}|}{I_{true}} X100\% \tag{3}$$

$$Percentage\ error = \frac{|26389 - 27180.67|}{27180.67} X100\% \tag{4}$$

$$Percentage\ error = 2.91\%.$$

In a controlled indoor setup, the robot, with a maximum speed of 0.5 m/s (1.8 km/h), was equipped with infrared (IR) sensors to detect left- or rightward lane deviations along a predefined track. The experiment was conducted in a 10 m × 10 m room, where the path formed a closed loop consisting of four laps. The track design included turns with a minimum angle of 30°, enabling smooth right- and left-hand maneuvers (Fig. 14).

The actual trajectory was compared with a reference (error-free) path to measure deviation. The difference was calculated as the **average displacement per point**, with the **relative error** defined as:

$$E = P_original + \Delta P, \tag{5}$$

Where:

ΔP- represents the introduced deviation,

The error was computed for each coordinate $P(x, y)$ along the original path.

The analysis showed a **relative error of 1.6 %**, meaning that the car's path deviated by no more than **±0.16 m**, or **±1.6 % of the total room dimension**.

4. Discussion

Sound-based indoor navigation systems offer visually impaired users a means to perceive and move through their environment by translating environmental information into auditory cues. These solutions are generally cost-effective and straightforward to implement, yet they depend on users accurately interpreting the presented sounds (Fig. 15). The present study sought to create an affordable, robust sound navigation system that enhances independence and quality of life for visually impaired individuals, prioritizing a design that is easy to configure, simple to install, economical, and energy-efficient. The system functions offline, employs voice commands for navigation, and achieves high predictive accuracy. As indicated by the normalized confusion matrix in Table 3, the approach attains a strong accuracy rate of 91%, Reflecting its effectiveness in supporting visual impairment. Through extensive testing, the system demonstrates reliable performance and offers practical solutions for indoor wayfinding. The prototype's combination of simplicity, affordability, and resilience positions it as a valuable tool for routine navigation tasks, enabling visually impaired users to attain greater autonomy and safety indoors. The vehicle supporting this system features a compact, efficient design. Equipped with an N20 DC gear motor and specified parameters, it can reach a maximum speed of 0.5 m/s under optimal conditions, operates at 80% Motor efficiency, and delivers 3 Nm of torque at full load. With dimensions of 40 × 30 × 30 cm and an approximate weight of 8 kg, the design contributes to stable movement. Powered by a 12-V motor, the vehicle is well-suited for indoor use and select controlled outdoor environments, enabling effective

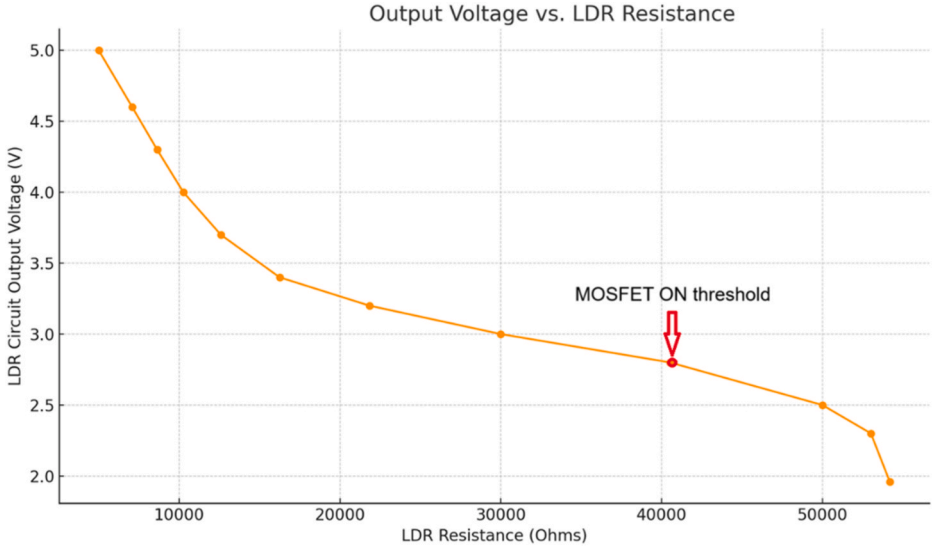


Fig. 12. The graph illustrates the relationship between light intensity (Lux) and the output voltage of the LDR-based voltage divider circuit. As light intensity increases, the LDR resistance decreases, resulting in a higher output voltage at the transistor gate. The red arrow indicates the MOSFET ON threshold at 2.8V; when the output voltage exceeds this threshold, the MOSFET is activated, allowing current to flow and turning on the robot's wheel.

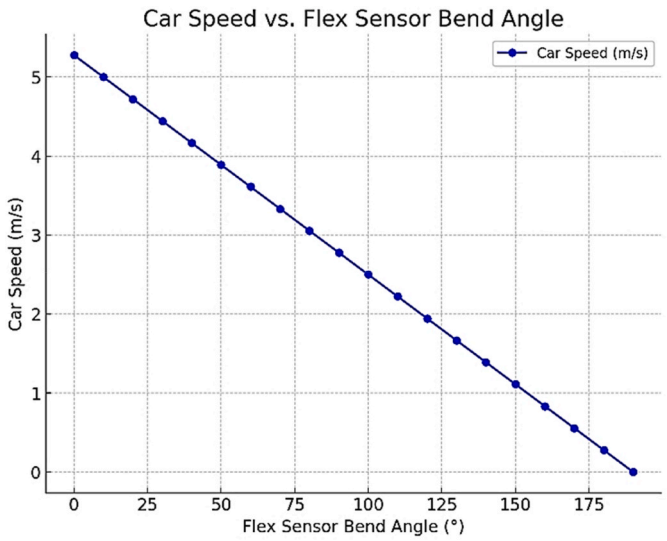


Fig. 13. Graph illustrating the relationship between the flex sensor's bending angle and the corresponding speed of the robot car.

navigation along pre-laid tracks or other managed pathways. Despite its strengths, several limitations remain, pointing to areas where further enhancements or adaptive solutions could improve usability and reliability for users with specific needs. Overall, this approach holds promise for increasing independence among visually impaired individuals and establishes a foundation for future advances in autonomous navigation technologies. Such developments could broaden mobility options, including support for train travel through route planning to ticket offices and boarding points, thereby improving accessibility within transportation systems.

The cited studies evaluate the performance and outcomes of diverse technologies developed to assist visually impaired individuals in navigating indoor environments (Table 4). Many prior solutions deploy complex systems that integrate advanced methods such as camera-based data processing, cyber-physical interfaces, text and door recognition, motion estimation, and augmented reality, necessitating sophisticated algorithms and intricate data handling, including fusion of visual and inertial data and multi-sensor tracking. In contrast, the present study adopts a more straightforward approach that emphasizes ease of implementation, speed, and safety. Core features include light-sensor readings to detect and maintain alignment with pathways, mechanical compass data to ensure orientation and real-time alignment, and voice recognition to command vehicle movement along the route, thereby providing an intuitive user interface. The voice recognition component achieved 91.04% accuracy, with 97.14% precision and 87.18% Recall, indicating high reliability in responding to trained commands. Grounded in a light-dependent resistor circuit, the cart followed the light path with high accuracy, exhibiting a 4 % error rate primarily attributable to background noise affecting sensor readings. The speed-control system, which integrates flex sensors and an electronic circuit to regulate the main power supply, effectively managed the vehicle's speed, yielding a 3% Error rate due to electrical fluctuations and mechanical inconsistencies. This data-driven, direct approach offers clear advantages over other commercial systems (Table 4) by reducing much of the complexity associated with those systems. Significant benefits include:

- o Safety: Indoor environments are typically not crowded, reducing the risk of external interference.
- o Low Cost: The approach utilizes affordable hardware and technologies, making it accessible to a broader range of users.
- o Faster Response Time: System simplicity enables quicker processing and more efficient operation.
- o Ease of Implementation: The system is easier to install and maintain compared to more complex solutions.

The proposed system emphasizes practicality and cost-effectiveness, offering a viable alternative to more complex and expensive navigation tools. This makes it particularly suitable for users who seek a straightforward and reliable indoor navigation aid.

5. Study limitations

1. The vehicle may not be suitable for individuals **with musculoskeletal problems**, as they **might** find it difficult to control. Alternative control mechanisms or adaptive devices **should be considered** to ensure accessibility for all users.
2. Light strips must be installed along the planned paths for the vehicle's navigation system to function optimally. **Additionally, these strips require regular cleaning to remove dust and dirt**, ensuring their effectiveness in guiding the vehicle's photodiodes.
3. The vehicle relies on battery power, which **necessitates** periodic charging and maintenance to **maintain functionality.** Battery degradation over time may **impact** the vehicle's performance and operational range.
4. The system assumes an obstacle-free environment along the planned path. **Unforeseen obstacles could interfere** with the vehicle's navigation, potentially **stopping or diverting** its movement.

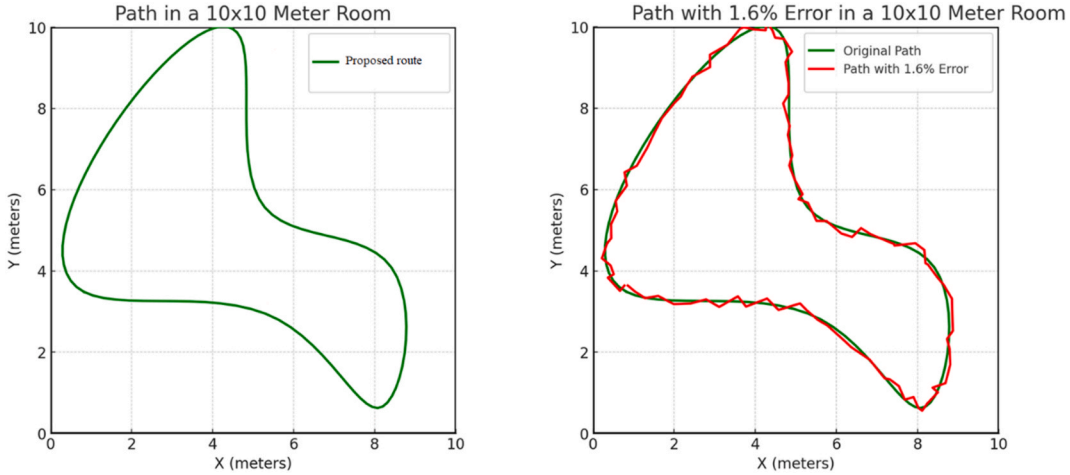


Fig. 14. Experimental analysis of robot car movement along a predefined path within a controlled 10 m × 10 m environment. The vehicle, equipped with infrared (IR) sensors, monitors path deviations, with a relative error of 1.6 % calculated at each point along the route.

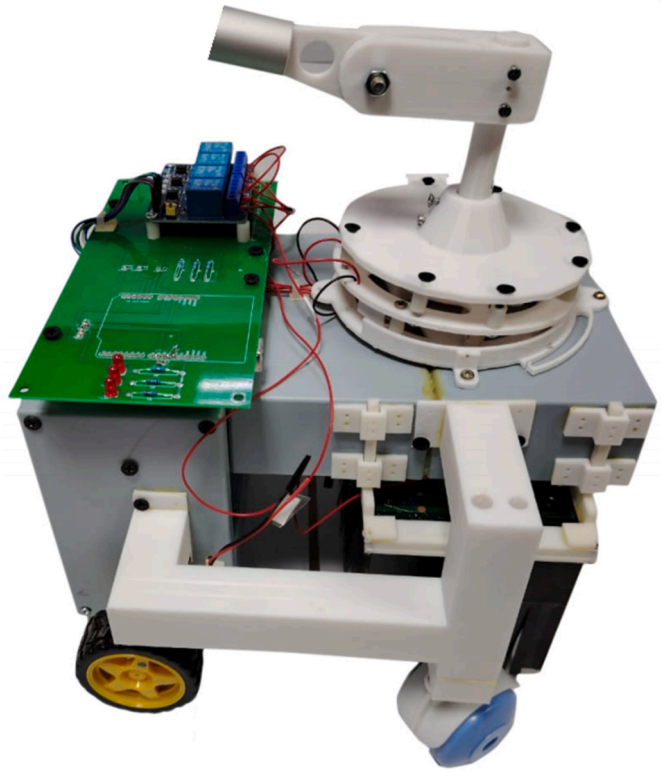


Fig. 15. The figure shows a prototype model of the proposed robot car, which comprises an electronic circuit for light tracking, a mechanical compass for user alignment, and a motor speed control system.

Regular **inspections of the path are necessary** to prevent such disruptions.

5. The vehicle's turning limitation is 90°, meaning it cannot rotate **fully** around itself. This limitation **stems from the navigation system** and should be considered when designing routes or determining turning spaces, as the vehicle may need to stop or reorient before completing a full turn.
6. The navigation experiment is a straightforward test to assess the vehicle's navigation error rate. It was conducted in a **controlled environment** and may not be fully generalizable to all users. Variations in user experience or physical conditions could affect performance in real-world settings. Future evaluations involving **diverse users** are recommended to refine and optimize system performance.
7. Most of the sensors employed in this study were **custom-built and innovative**, including the **mechanical compass, light sensor, and vibration motors**, as shown in Figs. 6, 9 and 10. The electronic components used in fabricating these sensors, such as the **LDR** and **flex sensors**, have also been verified for functionality and reliability.
8. It should be noted that **sensor performance tends to degrade under intense lighting conditions**, such as those found in media studios, exhibition halls, and entertainment venues (e.g., clubs), as well as in **noisy environments** like football stadiums and shopping malls. Therefore, the current prototype is primarily designed for **home and quiet indoor applications.**
9. The authors are currently pursuing **further research** to integrate IoT technologies into facilities serving people with disabilities in Saudi Arabia. These ongoing projects aim to advance indoor navigation, real-time tracking, and assistive systems to foster **safer, more accessible environments** for this essential segment of society.

6. Conclusion

In developed countries, assisting people with special needs, such as

Table 4
Comparison between some indoor navigation systems.

Authors	Technique	Method	System evaluation
Plikynaset al. (Plikynas et al., 2020)	Evaluation criteria prioritized by blind experts guide the development of indoor navigation systems.	Mapping the Features of Existing Technologies to User Needs	The qualitative value is SEp = 4
Xiao, J. et al. (Xiao et al., 2015)	A human-in-the-loop cyber-physical system that interprets semantic entities by interacting with both the physical world and the cyber domain.	A Navigation Assistance Framework for the Visually Impaired	The qualitative value is SEp = 3.13
Mekhalfi et al. (Mekhalfi et al., 2016)	Camera, IMU, and laser sensors	Restoring sight for individuals who are blind in indoor environments.	Qualitative value is SEp = 3.03
Tian et al. (Tian et al., 2013)	Text recognition and door detection	Towards a computer vision-based wayfinding aid for blind individuals navigating unfamiliar indoor environments.	Accuracy Medium
Lee and Medioni (Lee & Medioni, 2016)	Camera motion estimation	Wearable navigation system for the visually impaired based on RGB-D camera technology. Computer Vision and Image Understanding.	Accuracy Medium
Manlises et al. (Manlises et al., 2016)	CAMShift tracking	Indoor navigation system for the visually impaired based on computer vision, utilizing the CAMShift and D* algorithms.	Medium
Kanwal et al. (Kanwal et al., 2015)	Corner detection using visual and inertial data	A navigation system for the visually impaired: integrating vision and depth sensors.	Good
B Li et al. (Li et al., 2018)	Obstacle detection camera motion estimation by combining visual and inertial data	Vision-based mobile indoor navigation aid for blind individuals.	Good
Legge et al. (Legge et al., 2013)	Digital tag recognition	Indoor navigation for individuals with visual impairment using a digital signage system.	Medium
Zeb et al. (Zeb et al., 2014)	AR marker recognition	Vision-based indoor auditory assistance for blind individuals in semi-controlled environments.	Medium
Cheraghi SA et al. (Cheraghi et al., 2017)	Proximity detection, IMU	Beacon-based indoor wayfinding system for the blind, visually impaired, and disoriented individuals.	Good
Kunhoth J. et al. (Kunhoth et al., 2019)	CNN-based scene recognition, QR code recognition, and BLE-based positioning	Comparative analysis of indoor navigation systems for people with visual impairments: Computer vision vs. BLE technology.	The average error in BLE is 4.3 steps.

the blind, remains a critical priority, and automated vehicle control systems have proven practical by enabling navigation along pre-laid, light-sourced track strips with real-time guidance and ensuring orientation via microcontrollers, photodiodes, and electromagnets that drive the wheels.

A comprehensive electrical circuit for light sensing has been developed to accommodate the floor setup, with Python and Arduino used to program and control the robotic car for reliable operation. The introduction of such an automatic tracking system affords visually impaired individuals increased autonomy and safety in indoor spaces.

An integrated sound-based tracking and localization prototype was also designed to enhance indoor navigation, built around a Raspberry Pi 4, Arduino Uno, light-tracking sensors, a mechanical compass, and a headset, offering a seamless user experience with voice commands; the prototype employs an LSTM model trained to recognize specific commands while hardware is controlled offline via Python and C++ over a Wi-Fi hotspot.

7. Suggestions for improvement and future work

- Pilot trials with 5–10 visually impaired participants, conducted in collaboration with local rehabilitation centers or assistive technology organizations,
- Usability feedback collected through structured interviews and System Usability Scale (SUS) questionnaires,
- Iterative design refinements based on observed challenges and user preferences to strengthen practical value, identify edge cases, and ensure alignment with user needs.
- The current system suits small indoor spaces, and scaling to larger areas will require dynamic path planning, modular light extensions, and centralized or edge-based real-time control.
- We propose enhancements including adaptive lighting, infrared strips, and sensor fusion to improve accuracy in dusty or highly variable ambient-light environments.
- Future versions may adopt wearable haptic bands or belts to reduce hand fatigue and broaden usability.
- To improve performance in noisy or unstable lighting conditions, suggested developments include infrared reflective strips, electromagnetic sensors, and a voice-isolating wearable mask with a directional microphone for robustness and scalability.
- Future improvements will expand the voice recognition system by training on a more diverse dataset—encompassing various accents, environments, and additional commands—while incorporating noise filtering techniques to bolster resilience against ambient noise.
- To enhance user adaptability, upcoming versions will feature tutorial modes, adjustable vibration feedback, and alternative input methods such as buttons and mobile apps; although initial tests occurred in controlled settings, real-world validation with visually impaired users is planned.
- These trials will assess usability, comfort, and trust in autonomous navigation, ensuring broader accessibility and improved performance across diverse user needs and real-world indoor environments.

CRediT authorship contribution statement

Emad Malaekah: Writing – original draft, Methodology, Conceptualization. **Othman Alfahad:** Formal analysis. **Mohsen Bakouri:** Funding acquisition. **Adel Gadallah:** Formal analysis. **Suvad Selman:** Investigation. **Ahmed Al Rashdi:** Validation. **Husham Saied:** Software, Conceptualization.

Declaration

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Data availability statement

Data Availability Statements are available in the section "**Supplementary Materials**" at https://figshare.com/s/55e45e13c65b69faa793.

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Conflicts of interest

The authors declare that they have no conflicts of interest.

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Supplementary materials

The following supporting information can be downloaded at https://figshare.com/s/55e45e13c65b69faa793.

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Glossary

The following abbreviations are used in this manuscript:

AMD: Age-related Macular Degeneration
LED: Light Emitting Diode
BLE: Bluetooth Low Energy
LDR: Light-Dependent Resistor
AI: Artificial intelligence
GPS: Global Positioning System
MFCC: Mel-frequency cepstral coefficients
FIR: Finite Impulse Response
RSSI: Received Signal Strength Indicator
MAC: Media Access Control
dBm: decibel milliwatts
DC: Direct Current
MOSFET: Metal-Oxide-Semiconductor Field-Effect Transistor
V_GS: Gate-Source voltage